

Probability Theory and Statistics

Lecture 11

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Normal Distribution

Definition

A continuous random variable Y has a *normal distribution* with parameters $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ if its density is

$$f_Y(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R}.$$

Notation: $X \sim N(\mu; \sigma^2)$. $N(0; 1)$ is called *standard normal*.

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Standardizing the normal: if $Y \sim N(\mu; \sigma^2)$, then

$$\frac{Y - \mathbb{E}(Y)}{\mathbb{D}(Y)} \sim N(0; 1).$$

Behavior of the Sample Mean

In general: let $X_1, X_2, \dots, X_n$ be independent, **identically distributed** random variables with $\mathbb{E}(X_i^2) < \infty$.

(Identically distributed = they all have the same distribution function.)

Abbreviation: i.i.d. = independent and identically distributed.

Behavior of the Sample Mean

In general: let $X_1, X_2, \dots, X_n$ be independent, **identically distributed** random variables with $\mathbb{E}(X_i^2) < \infty$.

Let $S_n = \sum_{i=1}^n X_i$. Then

$$\mathbb{E}(S_n) = \mathbb{E}\left(\sum_{i=1}^n X_i\right) \stackrel{\text{linearity of } \mathbb{E}}{=} \sum_{i=1}^n \mathbb{E}(X_i) \stackrel{\text{i.i.d.}}{=} n\mathbb{E}(X_1).$$

$$\mathbb{D}^2(S_n) = \mathbb{D}^2\left(\sum_{i=1}^n X_i\right) \stackrel{\text{indep.}}{=} \sum_{i=1}^n \mathbb{D}^2(X_i) \stackrel{\text{i.i.d.}}{=} n\mathbb{D}^2(X_1).$$

Consider the **sample mean** $\overline{X}_n = \frac{S_n}{n}$. We see: $\mathbb{E}(\overline{X}_n) = \mathbb{E}(X_1)$, and

$$\mathbb{D}(\overline{X}_n) = \mathbb{D}(S_n/n) = \frac{1}{n}\mathbb{D}(S_n) = \frac{\mathbb{D}(X_1)}{\sqrt{n}} \longrightarrow 0.$$

Convergence in Distribution

Definition

Let $X_1, X_2, \dots$ be a sequence of random variables with distribution functions F_{X_n} .

We say that this sequence converges in distribution to a random variable Z with distribution function F_Z if, as $n \rightarrow \infty$,

$$F_{X_n}(x) \rightarrow F_Z(x)$$

for every $x \in \mathbb{R}$ at which F_Z is continuous.

Notation: $X_n \xrightarrow{d} Z$.

Thus the de Moivre–Laplace theorem asserts convergence in distribution. Historically the first central limit theorem.

A special case of the more general central limit theorem (CLT) coming soon.

Convergence in Distribution, Example 1: Normal with Variance Tending to Zero

For $n \in \mathbb{N}$, let X_n be normal with mean 0 and variance $1/n$.

We saw: as the standard deviation tends to 0 (i.e., $n \rightarrow \infty$), the density becomes narrower and more peaked.

For the distribution functions we have:

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = \lim_{n \rightarrow \infty} \mathbb{P}(X_n < x) = 0 \quad \forall x < 0, \quad \lim_{n \rightarrow \infty} F_{X_n}(x) = 1 \quad \forall x > 0.$$

Hence, except at $x = 0$, $F_{X_n}(x)$ tends to

$$F(x) = \begin{cases} 0, & \text{if } x \leq 0, \\ 1, & \text{if } x > 0. \end{cases}$$

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This is the [Heaviside function](#) (left-continuous version).

Note: F is also a distribution function—that of the constant-0 random variable with probability 1. Therefore X_n converges in distribution to the constant-0 random variable X .

Convergence in Distribution, Example 2: Poisson Approximation of the Binomial

Let $\lambda > 0$, and for $n \in \mathbb{N}$ let $X_n \sim B(n, \lambda/n)$. Also let $X \sim \text{Pois}(\lambda)$.

Earlier we saw: for every $k \in \mathbb{N}$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k) = \mathbb{P}(X = k). \quad (1)$$

Since the ranges of X_n and X are subsets of $\{0, 1, 2, \dots\}$, the functions F_{X_n} and F_X are constant on $(-\infty, 0]$, $(0, 1]$, $(1, 2]$, $(2, 3]$, etc.

Therefore convergence in distribution follows from (1) (clear? It suffices to show $F_{X_n}(k) \rightarrow F_X(k)$ for all $k \in \{0, 1, \dots\}$. Although F_X is not continuous at these points, if convergence holds at those points, then by piecewise constancy it holds everywhere.)

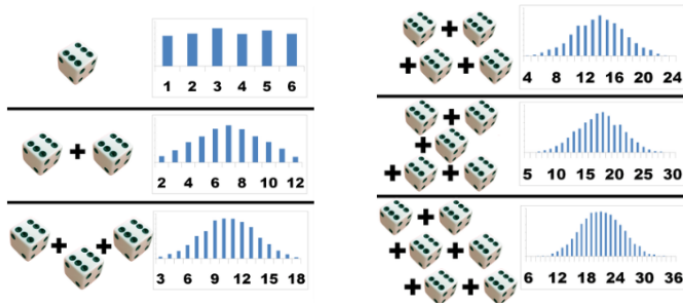
Here discrete random variables converge in distribution to another discrete one.

Central Limit Theorem (CLT)

Theorem

Let $X_1, X_2, \dots$ be independent, identically distributed random variables. Assume $\mathbb{E}(X_i^2) < \infty$ (finite variance), and let $Z \sim N(0; 1)$. Then

$$\frac{X_1 + \dots + X_n - n\mathbb{E}(X_1)}{\sqrt{n\mathbb{D}(X_1)}} \xrightarrow{d} Z \quad \text{as } n \rightarrow \infty.$$



A proof will follow soon.

Central Limit Theorem (CLT)

$$\frac{X_1 + \cdots + X_n - n\mathbb{E}(X_1)}{\sqrt{n}\mathbb{D}(X_1)} \xrightarrow{d} Z \sim N(0; 1).$$

Question: How can this be written as convergence of probabilities?

$$\mathbb{P}\left(\underbrace{\frac{X_1 + \cdots + X_n - n\mathbb{E}(X_1)}{\sqrt{n}\mathbb{D}(X_1)}}_{\substack{= \frac{\overline{X}_n - \mathbb{E}(\overline{X}_n)}{\mathbb{D}(\overline{X}_n)}, \\ \text{since } \mathbb{D}^2(\overline{X}_n) = \frac{n}{n^2} \mathbb{D}^2(X_1)}} < a\right) \xrightarrow{n \rightarrow \infty} \Phi(a), \quad a \in \mathbb{R}.$$

The de Moivre–Laplace theorem follows from the CLT: if $X_1, \dots, X_n$ are independent indicators of an event with probability p , then $X_1 + \cdots + X_n \sim B(n; p)$.

Example: CLT

A commercial radio station repeats 100 songs, choosing each next song independently of the past, uniformly from the 100 (so the same song may occur several times in a row). One of the 100 songs is Angela's favorite. Angela turns on the radio and listens until she has heard her favorite song in full for the 100th time, then she turns the radio off.

- 1 Let X be the number of songs Angela listens to until the **first** time she hears her favorite in full (so including the first occurrence of the favorite). What is the distribution of X and with what parameter(s)?
You do not need to justify the distribution type.
- 2 Using a suitable approximation, estimate the probability that Angela listens to at least 10 200 songs before turning the radio off.

Random Vectors

So far: we dealt with the distribution of a single continuous random variable: distribution function, density, expectation, distribution transforms, etc.

Now: the **joint** distribution of **several** continuous random variables (analogous to the discrete joint case) $\rightarrow$ **random vectors**.

*Do not be scared by the n -dimensional definitions/statements below. Sometimes we will need the general n case, but in practice most often **several**=2, and the two-dimensional case is simpler.*

Definition

Let $X_1, \dots, X_n$ be random variables defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$ for some positive integer n . Then the map

$$\underline{X} = (X_1, \dots, X_n): \Omega \rightarrow \mathbb{R}^n$$

*is called a **random vector**.*

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is called a *random vector*.

Definition

The *(joint) distribution function* of $\underline{X}$ is the

$$F_{\underline{X}}: \mathbb{R}^n \rightarrow [0, 1], \quad F_{\underline{X}}(x_1, \dots, x_n) = \mathbb{P}(X_1 < x_1, \dots, X_n < x_n),$$

a scalar-valued function of a vector argument.

Properties of the Joint Distribution Function

Reminder: for one random variable X , the distribution function F_X satisfies:

- $\lim_{x \rightarrow -\infty} F_X(x) = 0$,
- $\lim_{x \rightarrow \infty} F_X(x) = 1$,
- F_X is left-continuous, i.e., for every $x_1 \in \mathbb{R}$, $\lim_{x \uparrow x_1} F_X(x) = F_X(x_1)$,
- F_X is nondecreasing: if $x, y \in \mathbb{R}$ and $x \leq y$, then $F_X(x) \leq F_X(y)$.

What are the analogues of these properties for a random vector?

Properties of the Joint Distribution Function

If $F_{\underline{X}}$ is the joint distribution function of a random vector $\underline{X}$, then

- $\lim_{x_1 \rightarrow -\infty, x_2 \rightarrow -\infty, \dots, x_n \rightarrow -\infty} F_{\underline{X}}(x_1, \dots, x_n) = 0,$
- $\lim_{x_1 \rightarrow \infty, x_2 \rightarrow \infty, \dots, x_n \rightarrow \infty} F_{\underline{X}}(x_1, \dots, x_n) = 1,$
- $F_{\underline{X}}$ is “(partially) left-continuous in each variable”, i.e., for any $i \in \{1, \dots, n\}$, any $x_i \in \mathbb{R}$, and fixed $x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n \in \mathbb{R}$,

$$\lim_{x \uparrow x_i} F_{\underline{X}}(x_1, \dots, x_{i-1}, x, x_{i+1}, \dots, x_n) = F_{\underline{X}}(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n).$$

- If all but one variables are fixed, then it is (partially) nondecreasing. Namely, for $i \in \{1, \dots, n\}$ and given $x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n \in \mathbb{R}$, for any $x \leq y$,

$$F_{\underline{X}}(x_1, \dots, x_{i-1}, x, x_{i+1}, \dots, x_n) \leq F_{\underline{X}}(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n).$$

Properties of the Joint Distribution Function

Discrete case $\rightarrow$ joint probability table, marginals.

$Y \backslash X$	0	1	p_Y
0	$\frac{1}{4}$	0	$\frac{1}{4}$
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
2	0	$\frac{1}{4}$	$\frac{1}{4}$
p_X	$\frac{1}{2}$	$\frac{1}{2}$	1

In n dimensions it is the same, just with an n -dimensional joint mass function (harder to draw a table).

To obtain the marginal of one variable, **sum over all** the other variables.

Continuous Case?

Continuous case $\rightarrow$ what do we even mean by “continuous” here?

Example ($n = 2$): On the plane, choose a point uniformly at random on the segment of the line $y = x$ between $(0, 0)$ and $(1, 1)$; denote it by (X_1, X_2) .

Clearly $\mathbb{P}(X_2 = X_1) = 1$. Then X_1 and X_2 are each uniform on $[0, 1]$, so individually X_1 and X_2 are continuous random variables. However, the range of the random vector (X_1, X_2) is a subset of the one-dimensional subspace $\{(x, y) \in \mathbb{R}^2 : y = x\} \leq \mathbb{R}^2$.

We will see: (X_1, X_2) is not a **jointly continuous** random vector.

A **jointly continuous** 2-dimensional random vector (X, Y) always has a range with positive area.

It has a **joint density** $f_{(X,Y)}: \mathbb{R}^2 \rightarrow [0, \infty)$ whose double integral over all of $\mathbb{R}^2$ is 1.

Joint Density

Definition

Let $\underline{X} = (X_1, \dots, X_n)$ be a random vector. A function $f_{\underline{X}}: \mathbb{R}^n \rightarrow [0, \infty)$ is called the (joint) *density* of $\underline{X}$ if $f_{\underline{X}}$ has an improper Riemann integral over $\mathbb{R}^n$, and

$$\int_{-\infty}^{x_n} \cdots \int_{-\infty}^{x_1} f_{\underline{X}}(z_1, \dots, z_n) \, dz_1 \cdots dz_n = F_{\underline{X}}(x_1, \dots, x_n) \\ = \mathbb{P}(X_1 < x_1, \dots, X_n < x_n).$$

We call X continuous if it has a joint density.

Remarks:

- In one dimension this is the same notion as continuity of the distribution.
- For a 2D vector $X = (X_1, X_2)$ we often write f_{X_1, X_2} instead of $f_X = f_{(X_1, X_2)}$.

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Simplest example: choose a point $Y = (Y_1, Y_2)$ uniformly at random in the open unit square $(0, 1)^2 \subset \mathbb{R}^2$. Its density is

$$f_Y(y_1, y_2) = \begin{cases} 1, & \text{if } 0 < y_1 < 1 \text{ and } 0 < y_2 < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Joint Density

Theorem

Let $\underline{X} = (X_1, \dots, X_n)$ be a random vector. If X is continuous, then the function

$$f_X(x_1, \dots, x_n) = \begin{cases} \partial_{x_1} \cdots \partial_{x_n} F_X(x_1, \dots, x_n), & \text{if this exists,} \\ 0, & \text{otherwise,} \end{cases}$$

is a density of X . (The partial derivatives may be taken in any order.)

Remark: Unlike in the one-variable statement, here the existence of a density (i.e., joint continuity of X) is an assumption, not a consequence.

Recall the earlier example with $X_2 = X_1$: $F_{(X_1, X_2)}$ is twice (totally) differentiable except along a few lines, yet $\partial_{x_1} \partial_{x_2} F_{(X_1, X_2)}(x_1, x_2)$ is 0 wherever it exists, which cannot be a joint density. (Clear?)

Thus (X_1, X_2) is not continuous, even though the marginals of X_1 and X_2 are continuous.

Characterization of joint densities:

Theorem

Let $f: \mathbb{R}^n \rightarrow [0, \infty)$. Then

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(z_1, \dots, z_n) \, dz_1 \cdots dz_n = 1$$

if and only if there exists a random vector X whose density is f .

Joint Density

In one dimension we used the density to compute, e.g., probabilities of the form $\mathbb{P}(a < X < b)$. In higher dimensions:

Theorem

Let $H \subseteq \mathbb{R}^n$ be Jordan-measurable and let $X = (X_1, \dots, X_n)$ be a continuous random vector. Then

$$\mathbb{P}(X \in H) = \int_H f_X(x) dx.$$

E.g., in $d = 2$ dimensions, for a random vector $X = (X_1, X_2)$ and a rectangle $H = (a, b) \times (c, d)$,

$$\begin{aligned} \mathbb{P}((X_1, X_2) \in H) &= \mathbb{P}(a < X_1 < b, c < X_2 < d) \\ &= \int_a^b \int_c^d f_{(X_1, X_2)}(x_1, x_2) dx_2 dx_1 \left(= \int_H f_X(x) dx \right). \end{aligned}$$

Example

Let $f: \mathbb{R}^2 \rightarrow [0, \infty)$ be

$$f(x, y) = \begin{cases} 4xy, & \text{if } 0 < x, y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- Does there exist a random vector (X, Y) with density f ?
- If yes: compute $\mathbb{P}(X < Y)$.

(Solution: yes; $1/2$; see the lecture.)

Example

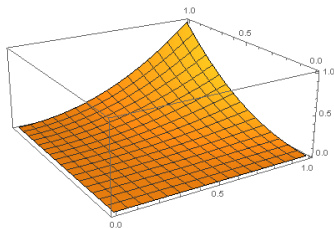
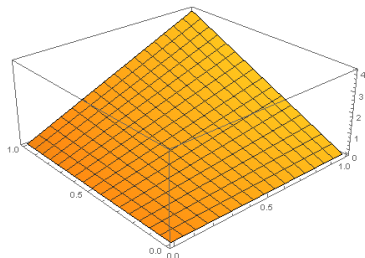
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- If yes: compute $\mathbb{P}(X < Y)$.

(Solution: yes; $1/2$; see the lecture.)

Joint density and distribution function in the example:



Marginal Distributions ($n = 2$)

On the next few slides we first consider the (less intimidating) $n = 2$ case. Afterwards we comment on general n .

If (X, Y) is a random vector, then F_X and F_Y are called the **marginal distribution functions**. How do we obtain them from $F_{(X,Y)}$?

In the joint distribution function, send the other coordinate to $+\infty$:

- $F_X(x) = \lim_{y \rightarrow \infty} F_{(X,Y)}(x, y), x \in \mathbb{R},$
- $F_Y(y) = \lim_{x \rightarrow \infty} F_{(X,Y)}(x, y), y \in \mathbb{R}.$

(Sketch of proof: see the lecture $\rightarrow$ continuity of probability measures!)

Marginal Distributions ($n = 2$)

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- $F_Y(y) = \lim_{x \rightarrow \infty} F_{(X,Y)}(x, y)$, $y \in \mathbb{R}$.

Question: how do we obtain the marginal density of X ?

- 1. approach: compute F_X and then differentiate.
- 2. approach: integrate out the joint density with respect to y :

$$f_X(x) = \int_{-\infty}^{\infty} f_{(X,Y)}(x, y) \, dy, \quad x \in \mathbb{R}.$$

Similarly, integrating with respect to x gives the marginal density of Y :

$$f_Y(y) = \int_{-\infty}^{\infty} f_{(X,Y)}(x, y) \, dx, \quad y \in \mathbb{R}.$$

Marginal Density ($n = 2$)

Example: choose a point $X = (X, Y)$ uniformly at random on the square $(0, 1)^2$:

$$f_X(x, y) = \begin{cases} 1, & \text{if } 0 < x < 1 \text{ and } 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Thus

$$f_X(x) = \int_{-\infty}^{\infty} f_X(x, y) \, dy = \int_0^1 1 \, dy = 1$$

for $x \in (0, 1)$ and 0 otherwise.

Similarly $f_Y(y) = 1$ for $y \in (0, 1)$ and 0 otherwise.

Both X and Y have uniform marginals on $(0, 1)$.

Marginal Density ($n = 2$)

$$f_X(x) = \int_{-\infty}^{\infty} f_{(X,Y)}(x,y) dy, \quad x \in \mathbb{R}.$$

Why is this true? $\rightarrow$ We saw:

$$\begin{aligned} F_X(x) &= \mathbb{P}(X < x) = \mathbb{P}(-\infty < X < x, -\infty < Y < \infty) = \\ &= \int_{-\infty}^x \int_{-\infty}^{\infty} f_{(X,Y)}(t,y) dy dt. \end{aligned}$$

Hence,

$$f_X(x) = F'_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy.$$

This explanation is valid if $F_{(X,Y)}$ is twice totally differentiable everywhere (hence all second-order mixed partials exist).

In the general case we obtain the same formula using substitution in integrals instead of differentiating.